

Ahmed Mishqat

+44 07881 598613 · Manchester, UK · mishqat0612@gmail.com · LinkedIn · GitHub · mishqat.github.io

PROFILE

Second-year BEng Mechanical Engineering student at the University of Manchester (First-Class Honours, Bicentenary Global Futures Scholar) with a broad engineering skillset spanning structural simulation, computational fluid dynamics, CAD design, robotics, and software development. Experienced in taking projects from concept through to physical prototype — with two IMechE competition wins this year as proof. Equally comfortable writing Python data-analysis pipelines, running FEA in Abaqus, or building and debugging embedded hardware at 2am the night before a deadline. Interested in roles and opportunities at the intersection of advanced engineering, simulation, and manufacturing.

EDUCATION

The University of Manchester — BEng Mechanical Engineering

2024 – 2026 (*Expected*)

- **Classification:** First-Class Honours (First Year)
- **Scholarship:** Bicentenary Global Futures Scholarship — the University's highest merit-based international award.
- **Relevant Modules:** Fluid Mechanics, Thermodynamics, Materials Science (microstructure, phase diagrams, deformation mechanisms, creep, weld metallurgy), Structures, Manufacturing Engineering, Mechanical Engineering Systems (ICE & Jet Engines)

Mastermind English Medium School — CAIE A Levels

2022 – 2024

- **Grades:** Physics (A*), Chemistry (A*), Mathematics (A*)
- **Awards:** Daily Star Award for Outstanding Academic Achievement; Intra-School Olympiad and Sports Awards

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks, Fusion 360 — Part & Assembly Modelling, Engineering Drawings, Design-for-Manufacture, Topological Optimisation
- **Structural FEA:** SolidWorks Simulation (stress & displacement studies, mesh convergence); Abaqus (nonlinear & contact analysis)
- **CFD:** Siemens Star-CCM+ — turbulence modelling, parametric mesh studies, post-processing and result interpretation
- **Materials & Characterisation:** Microstructure–property relationships, deformation and fracture mechanisms, creep, weld metallurgy; DBTT and impact energy data modelling
- **Workshop & Manufacturing:** Lathe & Milling, MIG/TIG Welding, 3D Printing, Bench Work, Assembly & Tolerancing
- **Programming & Data Analysis:** Python (NumPy, SciPy, Matplotlib), curve fitting, Monte Carlo uncertainty estimation, FFT-based signal analysis, GUI development (Tkinter)
- **Robotics & Mechatronics:** Arduino & Raspberry Pi, DC/Servo Motor Control, H-Bridge Drivers, Ultrasonic & IMU Sensing, Wireless Communication (Bluetooth, RF)
- **Web & Communication:** Static site development (HTML/CSS/JS, GitHub Pages); technical documentation, engineering drawing production, competition presentation

PROJECTS

IMechE Design Challenge 2026 — 1st Place, NW Regional Heat

Spring 2026

Autonomous Pipe-Climbing Robot — Advancing to National Finals

- **Concept & Design Leadership:** Drove the core mechanical architecture from ideation to competition build, establishing the clamping-and-climbing system that defined the team's design direction throughout.
- **Topological Optimisation & FEA:** Proposed and executed topology optimisation on primary structural components in SolidWorks, reducing part mass and material usage by over 30% — validated through iterative stress and displacement studies with mesh convergence checks.
- **Motor Selection & Torque Analysis:** Performed analytical torque and load calculations to specify drive and clamping motor requirements, ensuring reliable grip force and climbing performance across the full range of pipe diameters in scope.
- **Control & Sensor Fusion:** Authored the Arduino control architecture; when the ultrasonic module proved unreliable at extended range, developed a dead-reckoning algorithm from motor pulse timing that matched direct sensor accuracy.

IMechE CADathon 2026 — 2nd Place, Sustainability Category

Spring 2026

CrushIt: Gamified Pedal Can Crusher — SDG 11 · 8-Hour Design Sprint

- **Concept Origination:** Identified that aluminium cans are chronically under-recycled at high-footfall events not from apathy but from passive infrastructure — proposed CrushIt, a pedal-powered, gamified can crusher that makes recycling the fun option.
- **Incentivisation Design:** Developed a dual-sided competitive engagement mechanic that transformed a mundane waste action into a participatory experience — the key differentiator noted by the judging panel.
- **Mechanical Design & Materials:** Engineered the toggle linkage, pre-puncture spike platform, drainage system, and safety interlock; specified materials (304 SS, HDPE, galvanised mild steel, PP) with corrosion, fatigue, and recyclability justifications.
- **End-to-End Delivery:** Led SolidWorks CAD and full presentation deck within the 8-hour window, coordinating design intent, cost analysis, and SDG alignment narrative under competition time pressure.

Hack-A-Bot: Toxic Waste Disposal Rover, RoboSoc UoM

Spring 2025

24-Hour Robotics Hackathon

- Built a fully functional remotely controlled rover in 24 hours to transport a sensor-equipped payload across an obstacle course: DC-motor chassis, Arduino Uno with Bluetooth wireless control, and a servo-actuated claw-and-elevator lifting system with custom SolidWorks parts.
- Delivered an innovative claw-and-elevator mechanism under extreme time pressure; reinforced rapid systems integration, hardware debugging, and high-stakes team decision-making.

Robotic Arm Poker Dealer, RoboSoc UoM

Spring 2025

- Designed structural components for a servo-driven robotic arm in SolidWorks, 3D-printed and assembled custom parts, iterating hardware through progressive testing cycles.
- Identified and documented root cause of actuation failure (actuator-mass mismatch) as a structured engineering lesson in design-for-load.

Software & Portfolio Projects

2020 – Present

- Built and deployed a personal engineering portfolio website using HTML, CSS, JavaScript, and GitHub Pages — showcasing project writeups, CAD work, and technical documentation (mishqat.github.io).
- Developed Python GUI tools including a scientific calculator (NumPy), password manager, and text editor (Tkinter); practised algorithmic problem-solving through Codewars.

LEADERSHIP & EXTRA-CURRICULAR

President — Mathematics Club, Mastermind English Medium School

2023 – 2024

- Planned and directed Highschool Math-Fest, an annual competition with 70+ volunteers and 1,000+ participants; managed operations, logistics, and real-time problem-solving under pressure.